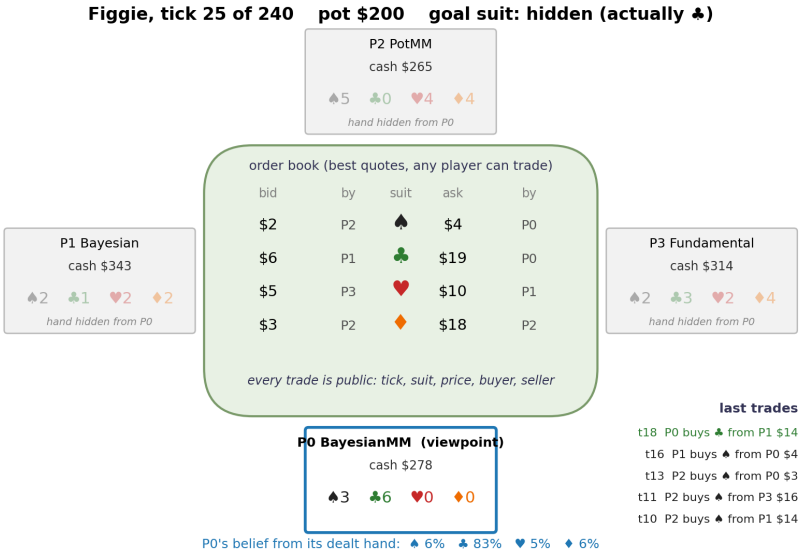
Reading the Order Flow: Information Extraction and Leakage in Learned Agents for the Trading Game Figgie

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1 Problem and motivation

In Figgie, 40 custom cards with no numbers or faces are dealt from one of 12 equally likely decks: one suit has 12 cards, one has 8 and two have 10. The goal suit is the **same color partner** of the 12-card suit (♠↔♣, ♥↔♦), i.e. if the deck is 12 ♠ / 10 ♣ / 10 ♥ / 8 ♦, the goal suit is ♣, and each ♣ pays \$10 plus a pot bonus to whoever holds the most. Players trade continuously, so a player who eagerly buys ♣ is likely to hold many clubs and believe ♣ is the goal. Existing approaches, such as `figment`'s Bayesian market maker, `Figgie-bot`'s exact inference, and the `CS224R PPO` agent [1], compute or approximate the goal-suit posterior given only their own hand. Instead, we seek to answer three questions:

1. How much does the public trade history **reveal** about the goal suit beyond a single hand, and how quickly over a 240-tick game?
2. Do agents with memory (GRU, transformer) **learn to infer** the goal suit despite only receiving rewards for profit?
3. How much does each agent's own action stream reveal or **leak** about its hand, and do the leakiest agents lose the most to a learned best response (exploiter)?

Elo and profit against a fixed field are the only metrics reported for Figgie agents, and they say nothing about **why** an agent wins, or how robust it is. Figgie is a clean testbed for a central open question in reinforcement learning under partial observability: does an agent trained only on reward learn to infer hidden state, and to infer it from *other agents' behaviour*? Strong agents for imperfect-information games such as DeepStack [12] and Pluribus [8] sidestep the question by computing beliefs explicitly; model-free agents with memory are expected to learn them implicitly, but this is rarely checked, because in poker or StarCraft the exact belief is unknown or intractable. In Figgie it is exact: the 12-deck posterior from any set of cards is a closed-form sum, and the simulator records the true deck and every hand. Figgie also has the multi-agent twist that every

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action is a signal: an agent that learns to read others' trades should also learn that its own trades are read, and exploitability, the standard robustness measure, has no closed form in a four-player general-sum game. The study exercises the core of the course: recurrent and transformer sequence models, representation learning via probes, and reinforcement learning with self-play.

2 Data

We use no external dataset. All data is generated by simulation in our engine, which implements the standard rules and the CS224R discretisation (240 ticks, 257 actions, prices 1–30). We have already replicated four public sources on this engine: the CS224R baselines and PPO pipeline [1], figment [2], Figgie-bot [3] and figgie-auto [4]. The figment and Figgie-bot ports reproduce every decision of the original code, and our PPO agent matches or exceeds every number in the CS224R report, including a 52% win rate (54% reported) against its strongest heuristic table. Each training or probing dataset is a set of simulated games with full logs: the deck, all hands, and every quote and trade.

3 Related work and relevance to the field

Market microstructure. Kyle [6] and Glosten–Milgrom [7] model how informed traders' orders reveal private information and how market makers protect themselves against it. Those models assume rational agents with known strategies. We measure revelation and leakage for learned and heuristic agents in a market where ground truth is exact, which is impossible on real exchanges.

Multi-agent learning. Exploitability is the standard robustness measure in two-player zero-sum games, but it has no closed form in multi-player general-sum games such as Figgie or six-player poker [8]. Learned best responses [9] are the practical substitute; we pair them with an explanation of *why* an agent is exploitable.

Interpretability of RL agents. Probing has shown that AlphaZero encodes human chess concepts [10], using linear probes [11]. Figgie offers an exact Bayesian target for the probe, so we can say not only whether a belief is encoded but how close it is to optimal.

4 Preliminary approach

4.1 Baselines

Belief: the exact hand-only Bayesian posterior over the 12 decks, which every published agent approximates. **Agents:** the CS224R PPO agent and the Bayesian market makers from figment and Figgie-bot, all replicated and verified on our engine; these are the reference points for strength, leakage and exploitability.

4.2 Method

Revelation bound. Train a supervised transformer that reads the public trade history up to tick t and predicts the goal suit. Compare its accuracy and log-loss over t with the exact hand-only posterior. The gap is an empirical upper bound on what any agent can gain from the order flow, measured separately against each population of opponents.

Extraction. Fit linear and small MLP probes from each learned agent's hidden state to (a) the exact hand-only posterior and (b) the true goal suit. Train new PPO agents whose observation includes the trade history, comparing a GRU, a transformer, and hand-crafted belief features, and track belief accuracy throughout training.

Leakage. For each agent, train a classifier that predicts that agent's hand composition from its actions alone; its accuracy above the prior is the agent's leakage.

Exploitability. For each agent, train a PPO best response at one seat with the other seats fixed, and estimate exploitability as the best response's profit gain over a no-information control. We decompose the gain into an information term (earlier correct goal-suit belief) and a pricing term, and test whether leakage predicts exploitability across agents. If time allows, we validate the estimator on small games where exact exploitability is computable (Kuhn and Leduc poker).

Expected results. (i) Goal-suit accuracy versus tick for hand-only, order-flow and probed agent beliefs; (ii) a table of Elo, profit, leakage and estimated exploitability for every agent; (iii) whether learned memory architectures close the gap to the revelation bound.

4.3 What we measure

Quantity	Definition
Revelation gap $\Delta_{\text{rev}}(t)$	Log-loss of the exact hand-only goal-suit posterior minus log-loss of the best predictor that also sees the public trade history up to tick t, in bits; reported for each opponent population.
Belief extraction $E_a(t)$	Log-loss of a probe on agent a's hidden state for the true goal suit, placed between the hand-only posterior ($E = 0$) and the revelation bound ($E = 1$).
Leakage L_a	Mutual information, in bits, between agent a's hand composition and its action stream, lower-bounded by the cross-entropy gain of a trained classifier over the prior.
Exploitability $\hat{\varepsilon}_a$	Profit per game of a learned best response at one seat against three copies of a, minus that of a no-information control. Decomposed into an information term and a pricing term.
Strength	Mean profit per game with 95% CIs, win rate and Elo over rotated mixed tables (already measured in the replication).

The central tests are: does $E_a > 0$ for any learned agent; does E_a increase with history-aware architectures (transformer $>$ GRU $>$ none); and is the rank correlation between L_a and $\hat{\varepsilon}_a$ positive across agents.

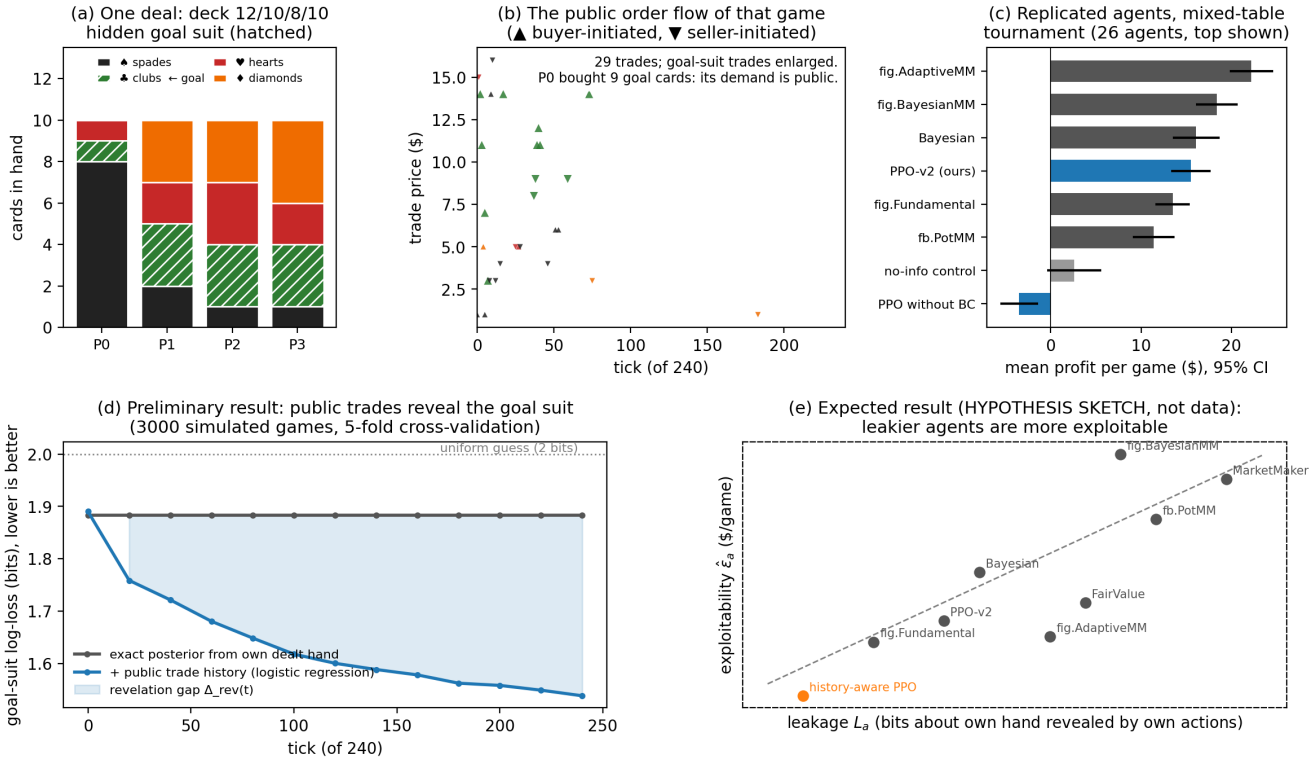


Figure 1: The setting and our first measurement, all on our engine with the replicated agents. (a) One deal: four hands of ten cards from a 12/10/8/10 deck; the goal suit (hatched) is hidden. (b) The public order flow of that game: every trade's tick, suit and price is visible to all players, and one player's heavy buying of the goal suit is plain to see. (c) Mean profit per game in a 26-agent mixed-table tournament of all replicated agents, including our reproduction of the CS224R PPO agent. (d) Preliminary result over 3,000 games: the exact posterior from a player's own dealt hand leaves 1.88 bits of uncertainty about the goal suit; a simple logistic regression that also reads the public trade history reduces this to 1.54 bits by the end of the game (top-1 accuracy 39% $\rightarrow$ 54%). This gap, which no published agent uses, is the information our project is about. (e) A *hypothesis sketch, not data*, of the relation we will test between how much an agent leaks and how exploitable it is.

4.4 What is new

- Prior Figgie work ranks agents only by profit or Elo against a fixed field: hand-written strategies in a discrete-event simulation [5], Bayesian and evolutionary bots [2, 3], and one PPO agent [1]. None measures what the order flow reveals or what an agent leaks.

- No published Figgie agent conditions its belief on opponents' trades; all use the player's own hand. We test whether learning closes that gap.
- No exploitability estimate exists for any Figgie agent, and none for a multi-player trading game that we are aware of.
- We release a single engine with four replicated sources whose decisions are verified against the original code, so future work can compare against the same agents.

Timeline. Midterm (3 weeks): replication results, revelation bound, first probe of PPO. Final (8 further weeks): history-aware PPO variants, leakage measurement, best-response exploitability and the leakage–exploitability analysis.

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